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Project Report

Parking Lot Management

Problem statement:

To build a “Parking Lot Management” system on a web-application that tracks entry and exit of vehicles in a parking lot, track available parking spaces, and calculate parking fees.

Aim:

To build a user-friendly, easily understandable, yet detailed parking lot management web-application using recently learnt modules of Figma (Prototype), Angular (Frontend), C# (API) and SQL (Backend).

The parking lot has a fixed number of slots (e.g., 50).

Users:

- Parking Attendant – records vehicle entry and exit.
- Supervisor – monitors parking usage and daily reports.

Prototyping:

For prototyping I have used technologies learnt from Figma and to create a simple layout of UI for workflow and design ideas.

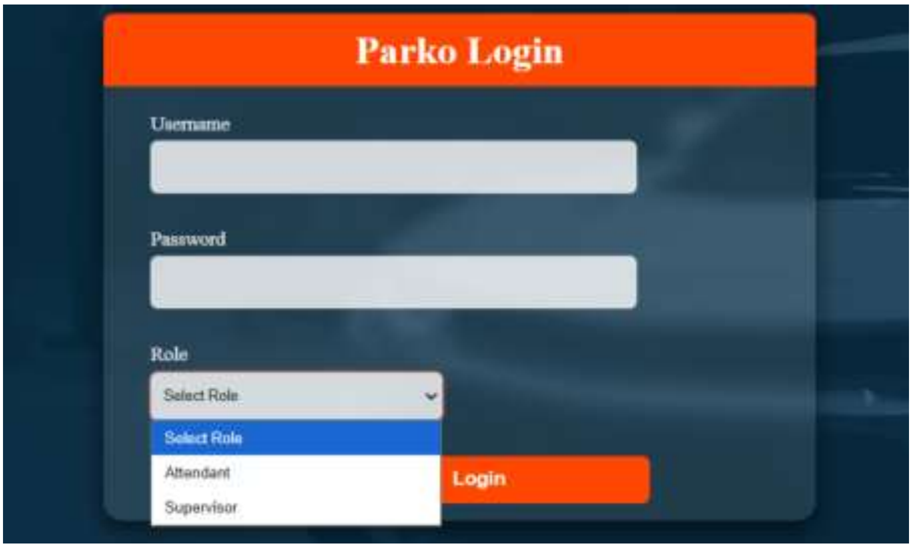
These design ideas and flow used in prototyping were used only for a reference and were modified during implementation.

For the primary colours, I have chosen orange and bluish-grey and other minimal colours to make the UI more attractive and catchier.

Frontend:

Frontend for the project was done using technologies learnt for Angular framework and the pages were designed using HTML, SCSS and Typescript.

- Login page:



The login page contains three fields; A username input field, password input field and a drop-down field.
The username and password fields are for authenticating any user with login credentials to access the web-application.
The drop-down field allows the user to login to their respective role page as either “Attendant” or “Supervisor”.

Only the user authenticated for either of the roles can access the web-application and in no case can a user cross login their role instead of assigned role.

- Attendant dashboard and attendant pages:

The attendant dashboard contains a simple dashboard with only two navigation buttons:

1. Entry a vehicle.
2. Exit a vehicle.

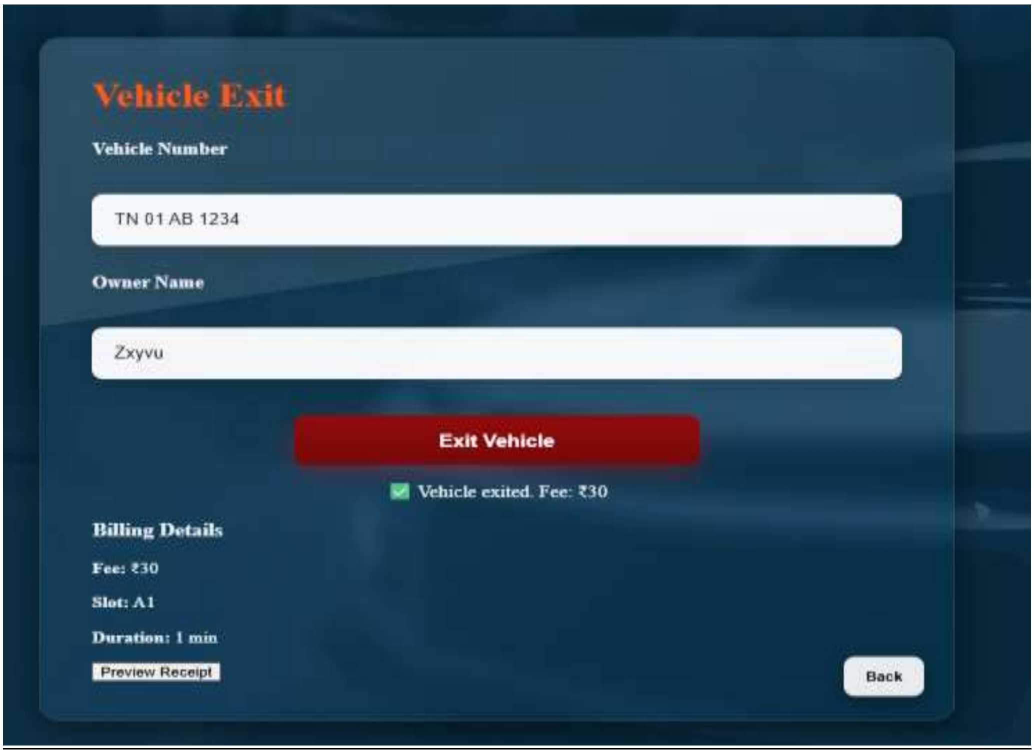
Both are designed like a floating card that can be used by the attendant to do his tasks.

1.Entry a vehicle:

The above page is the page for vehicle entry, where the attendant can input just two simple inputs "Vehicle number" and "Owner/Driver name".

After entering the details, the system will automatically assign you a nearest free slot available in the parking lot. The lot will be displayed just in the bottom of screen after entering the details. The owner/driver of vehicle can just park his car in the slot he's assigned.

2.Exit a vehicle



The exit vehicle page is almost as similar as entry vehicle page. When the owner/drive wants to exit the parking lot, the attendant has the enter the exact vehicle number and the exact owner/driver's name that was registered during entry.

Mismatch of any details won't allow the vehicle to pass through. This method prevents more theft and provides high security. For security reasons, it's impossible for the attendant to open the entry details.

Match of given details prompts a message appearing "Vehicle exited" and the slot will be freed for the next vehicles. Theres also a "preview receipt" button that opens a pop-up screen displaying the owner/driver name, vehicle number, In-time, Exit-

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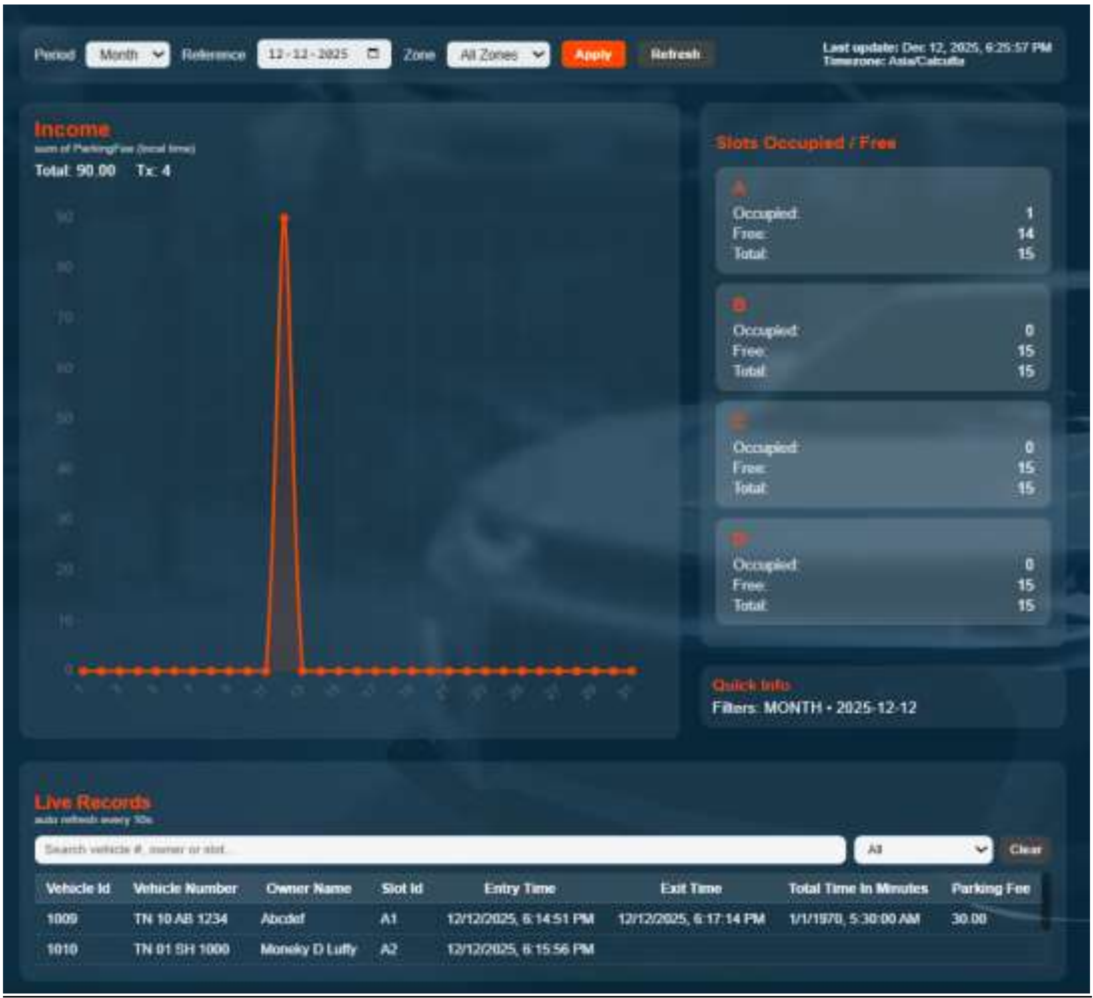
time, Total number of minutes the vehicle was parked and the final fee. Attendant can print the receipt if he or the drivers wants to do so.

- Supervisor dashboard:



The supervisor dashboard contains a detailed overview of history of vehicle record, current parking slot availability, and revenue graphs.

Filters can be used to sort out revenue for specified day, week, month or year. A table of records is assigned to view details of both exited vehicles from past for a safety record.

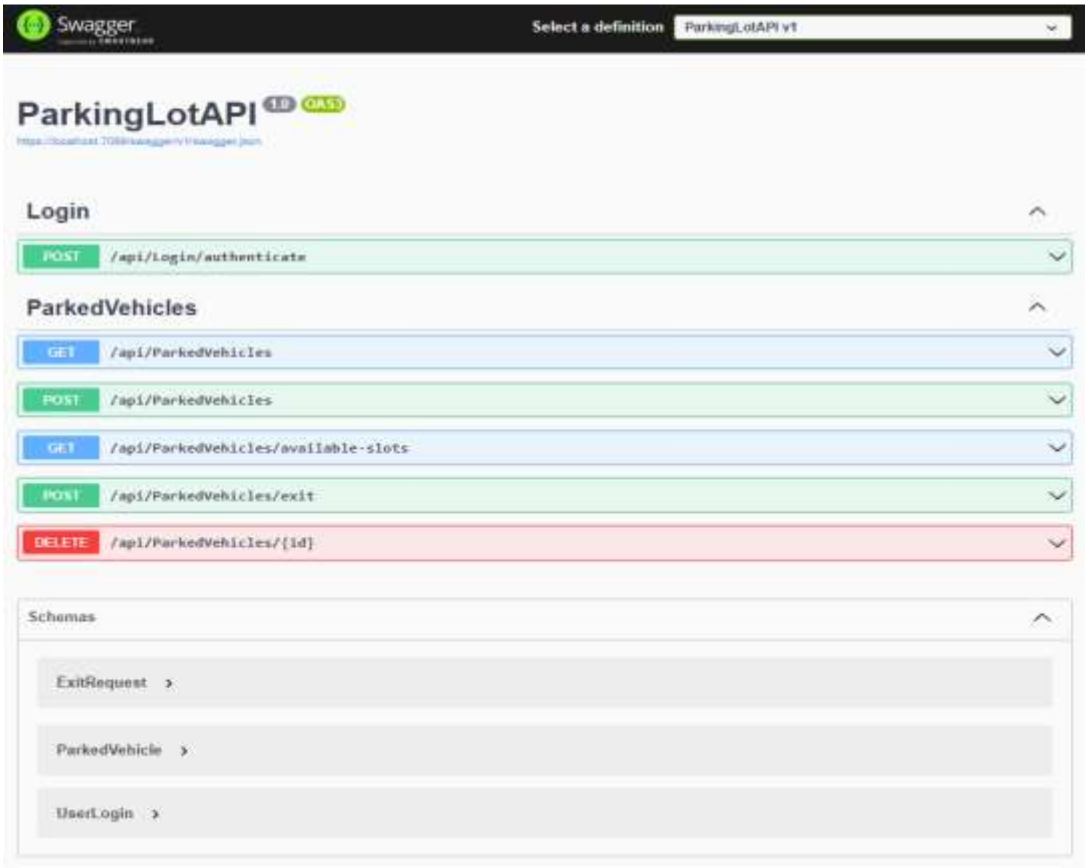
The graph updates on real time showing revenue obtained totally by the parking fee. The slots free or occupied displays live slots.

Just like how the parking attendant cannot view or edit the vehicle details, the supervisor cannot edit details of entry vehicles or entry/exit vehicles on their own.

API:

The API is a medium between the Frontend (Angular) and the Backend (SQL) and also used for some logics involved in assigning the parking slots and calculating the parking fee.

I used ASP.NET core for the API framework that uses C# for controllers and logics of code that interacts with backend.



The login controller path in API controls the login authentication and the roles.

The GET and POST API path for parked vehicles controls the vehicle entry as well as displays overall parking of both currently parked as well as exited (previous entries).

The GET available-slots path is the API call which provides the details of free slots in the parking lot.

The POST method for exit is used to exit the vehicles, frees the slot that vehicle was occupying and calculates parking fee.

Backend:

The backend is consisted of database in form of tables that contains all data like records of currently parked, previously parked and exited vehicles, free/occupied slots in parking lot, user credential database.

- Login DB:

Results Messages

	Id	Username	Password	Role
1	1	attendant1	1234	Attendant
2	2	supervisor1	5678	Supervisor
3	3	Attendant	attendant@123	Attendant
4	4	Supervisor	supervisor@123	Supervisor
5	5	attendant	attendant@123	Attendant
6	6	supervisor	supervisor@123	Supervisor

Contains the username, password and role credentials that were previously loaded into database. When the user tries to login, the API checks and authenticates for login only when all data from a row matches the input.

- Parked vehicle records:

82 % No issues found

Results Messages

	VehicleId	VehicleNumber	OwnerName	SlotId	EntryTime	ExitTime	TotalTimeInMinutes	ParkingFee
1	1009	TN 10 AB 1234	Abcdef	A1	2025-12-12 18:14:51.143	2025-12-12 18:17:14.983	2	30.00
2	1010	TN 01 SH 1000	Moneky D Luffy	A2	2025-12-12 18:15:56.133	NULL	NULL	NULL
3	1011	AA 11 AA 1111	Hari	A1	2025-12-12 18:19:19.360	2025-12-12 18:19:37.173	0	30.00
4	1012	TN 01 AB 1234	Zyxvu	A1	2025-12-12 18:22:47.597	2025-12-12 18:23:44.957	0	30.00

This table in the SQL database contains all details of vehicle, its number, the entry time etc. This table stores historical data and won't get deleted when the vehicle exits.

- Availability slot DB:

This tables shows list of slots from A1, A2… to D15 and their availability. This slot is used to determine if a new vehicle can be added in the entry or not and also used to determine the hierarchy for availability slot.

Results Messages

	Id	SlotId	IsOccupied
1	1	A1	0
2	2	A2	1
3	3	A3	0
4	4	A4	0
5	5	A5	0
6	6	A6	0
7	7	A7	0
8	8	A8	0
9	9	A9	0
10	10	A10	0
11	11	A11	0
12	12	A12	0
13	13	A13	0
14	14	A14	0

Demo result:

WhatsApp Video 2025-12-12 at 8.53.52

Conclusions:

Thus, the project- Parking Lot Management has been successfully developed as a web-application and tested to provide flawless user experience. The primary colour choices have only been used throughout the project to maintain a professional and clearer page with readable fonts and most importantly minimal components with maximum details.